Evaluation of Steganography

Classroom Presentation Speaking Script

1. Start with the introduction

You can say:

> “Good morning everyone. Today I am going to present my topic, **Evaluation of Steganography**.
>
> The reference paper I used is *Ancient and Modern Steganography*.
>
> Steganography means hiding information inside another medium so that the existence of the secret message is not easily noticed.
>
> Unlike cryptography, where the message is usually visible as encrypted data, steganography tries to hide the fact that communication is taking place.”

Then point to the **Prisoners Problem** figure in the paper and explain:

> “This figure gives a simple example. Alice wants to send a secret message to Bob, but Eve is watching the communication. The idea of steganography is to make the communication look normal to Eve.”

This is a very good opening because your classmates can understand the basic idea immediately.

2. Explain the difference between cryptography and steganography

Say:

> “There is an important difference between cryptography and steganography.
>
> Cryptography changes the message into an unreadable form, but the encrypted message can still attract attention.
>
> Steganography hides the message inside a normal-looking image, audio file, video or text.
>
> So the goal is not only to protect the message, but also to hide the existence of the message.”

Then ask your classmates:

> “For example, if you received a normal-looking image from a friend, would you immediately think there was a secret message inside it?”

That small question makes the presentation interactive.

3. Ancient steganography

This is one of the most interesting parts of your PDF.

Say:

> “Steganography is not a new technology. It was used even in ancient times.”

Example 1 - Wax tablet

Point to the ancient wax tablet image.

> “According to the paper, Demaratus used a wooden backing with a message and then covered it with wax. The message was hidden under the wax, so the communication was not obvious.”

Example 2 - Hidden messages

> “The paper also describes other historical examples, such as messages hidden using clothing, objects and physical materials.”

Example 3 - Geoglyphs

Point to the **Nazca Lines** image.

> “These large ground drawings are visible properly from the air. The paper discusses them as an example of information being hidden through a medium that does not reveal its meaning to an ordinary viewer.”

Example 4 - Cardan grille

Point to the Cardan grille figure.

> “Another interesting technique is the **Cardan grille**. A special grid is placed over ordinary text, and the selected positions reveal the hidden message.”

This section is where you can sound confident because you are explaining the history rather than technical formulas.

4. Transition from ancient to modern

Use this sentence:

> “So the basic idea has remained the same for hundreds of years - hide information inside something that looks ordinary. The difference is that modern technology allows us to do this digitally.”

That gives you a very natural transition.

5. Modern steganography

The paper identifies five main forms:

> “In the modern era, steganography can be applied to **text, images, audio, video and network protocols**.”

Text steganography:

> “The secret information is hidden inside text by techniques such as spaces, word shifting, abbreviations or selected characters.”

Image steganography:

> “A secret message is embedded into an image. This is one of the most common forms of digital steganography.”

Audio steganography:

> “The message is hidden inside an audio file. The technique takes advantage of the limitations of human hearing.”

Video steganography:

> “A video contains many image frames and audio, so it can provide more places to hide information.”

Protocol steganography:

> "Information can also be hidden inside network communication or protocol information."

These categories come directly from the PDF.

6. Spend the most time on image steganography

This should probably be the **main technical part** of your presentation.

Point to the flower/image steganography figure and say:

> "Here is the basic image steganography process. We start with a normal image, called the cover image. Then we have a secret message. An embedding algorithm combines the two and produces a stego image. Later, a decoding process extracts the secret message."

Then give a simple real-world explanation:

> "The important point is that the stego image should look almost identical to the original image."

7. Your practical coding demonstration

This is the part that can make your presentation much stronger than a normal theory presentation.

You can tell your classmates:

> "Now I will show a simple practical example of how image steganography works using Python."

Explain **LSB** in very simple language:

> "LSB stands for Least Significant Bit. A pixel has binary values. We can change the last bit of a pixel to store one bit of secret information. Because this is a very small change, the visual difference is usually difficult to notice."

Give a tiny example:

```
Original pixel: 10110110
Secret bit:            1
Modified pixel: 10110111
```

Then say:

> "Only the last bit has changed. That small change is enough to store information."

Do not spend too long explaining binary unless your teacher asks.

8. Show your actual image

This is where you should display your **carrier image** → **stego image** → **difference image**.

Say:

> "This is my original carrier image."

Then:

> "This is the image after embedding the secret message."

Then:

> "To our eyes, the two images look almost the same. But internally, some pixel bits have changed."

Then show the difference image:

> “This difference image makes those changed areas easier to see. It is not the normal image seen by the receiver; it is only useful for evaluating the changes.”

This makes your topic feel like a real experiment rather than just theory.

9. Explain how you evaluate the steganography

Because your title is **Evaluation of Steganography**, make sure you actually discuss evaluation.

You can say:

> “When we evaluate a steganography technique, we mainly ask three questions:
>
> 1. Can we hide enough data?
> 2. Does the image still look normal?
> 3. Can someone detect that a secret message is present?”

Then explain:

> **Security:**
> “How difficult is it for an attacker to detect the hidden information?”
>
> **Capacity:**
> “How much secret data can be hidden?”
>
> **Image quality:**
> “How much does the image change after embedding?”

This is the simplest way to connect your practical coding example to your presentation title.

10. Steganalysis

Now introduce the opposite side.

Say:

> “If steganography is about hiding information, **steganalysis** is about finding it.”

Then:

> “The paper explains several attack models, such as Stego-Only Attack, Known Cover Attack, Known Message Attack, Chosen Stego Attack and Chosen Message Attack.”

You do not need to memorize every definition. For a classroom presentation, explain the idea:

> “An attacker can compare images, study statistical properties, look for unusual patterns, or use specialized steganalysis tools.”

Then point to the histogram figure.

> “This figure shows histograms before and after hiding information. Statistical analysis can sometimes reveal differences caused by embedding.”

This is directly connected to the evaluation part.

11. Advantages and disadvantages

Use simple language.

Advantages

Say:

"The major advantage is that the communication does not obviously look secret. It can also be used for digital watermarking and protection of digital content."

Disadvantages

Then:

"The disadvantage is that if too much information is hidden, the media can become distorted and easier to detect. Steganography can also be misused for hiding malicious information."

The paper discusses both useful applications and misuse.

12. Conclusion

End with this:

"To conclude, steganography is an old idea with modern digital applications.

In ancient times, messages were hidden in physical objects, writing and other materials.

Today, information can be hidden in images, audio, video, text and network communication.

Image steganography, especially techniques such as LSB-based hiding, shows how a very small change can carry secret information.

At the same time, steganalysis is important because hidden information can sometimes be detected through statistical or other analysis.

So the effectiveness of steganography depends on security, capacity and maintaining the quality of the cover medium."

Then simply say:

"Thank you. I am ready for your questions."

Questions your classmates or teacher may ask

“What is the difference between steganography and cryptography?”

“Cryptography hides the meaning of the message, while steganography tries to hide the existence of the message itself.”

“Why is LSB used?”

“Because changing the least significant bit causes only a very small numerical change in a pixel, so the visual effect is usually small.”

“Can steganography be detected?”

“Yes. Steganalysis techniques can look for statistical or visual abnormalities in the stego medium.”

“Why use an image?”

“Images contain a large number of pixels, so they provide many values in which small changes can be made.”

“Is steganography completely secure?”

“No. It can improve secrecy, but it is not automatically secure. The technique, embedding capacity and detectability all matter.”

Presentation tip

Do not read the PDF line by line. Your flow should be: **What is steganography?** → **Ancient examples** → **Modern types** → **Image steganography** → **Your Python demo** → **Evaluation** → **Steganalysis** → **Advantages/disadvantages** → **Conclusion**.

Spend roughly half of your speaking time on the practical image + LSB demonstration, because that is the part most likely to make the topic feel like a real experiment.